

USING THE EFI EXTENDED SCSI PASS THRU PROTOCOL

This appendix describes how an EFI utility might gain access to the EFI SCSI Pass Thru interfaces. The basic concept is to use the *EFI_BOOT_SERVICES.LocateHandle()* boot service to retrieve the list of handles that support the *EFI_EXT_SCSI_PASS_THRU_PROTOCOL*. Each of these handles represents a different SCSI channel present in the system. Each of these handles can then be used to retrieve the *EFI_EXT_SCSI_PASS_THRU_PROTOCOL* interface with the *EFI_BOOT_SERVICES.HandleProtocol()* boot service. The *EFI_EXT_SCSI_PASS_THRU_PROTOCOL* interface provides the services required to access any of the SCSI devices attached to a SCSI channel. The services of the *EFI_EXT_SCSI_PASS_THRU_PROTOCOL* are then to loop through the Target IDs of all the SCSI devices on the SCSI channel.

```
#include "efi.h"
#include "efilib.h"

#include EFI_PROTOCOL_DEFINITION(ExtScsiPassThru)

EFI_GUID gEfiExtScsiPassThruProtocolGuid =
EFI_EXT_SCSI_PASS_THRU_PROTOCOL_GUID;

EFI_STATUS
UtilityEntryPoint(
    EFI_HANDLE ImageHandle,
    EFI_SYSTEM_TABLE SystemTable
)
{
    EFI_STATUS Status;
    UINTN      NoHandles;
    EFI_HANDLE *HandleBuffer;
    UINTN      Index;
    EFI_EXT_SCSI_PASS_THRU_PROTOCOL *ExtScsiPassThruProtocol;

    //
    // Initialize EFI Library
    //
    InitializeLib (ImageHandle, SystemTable);

    //
    // Get list of handles that support the
    // EFI_EXT_SCSI_PASS_THRU_PROTOCOL
    //
    NoHandles = 0;
    HandleBuffer = NULL;
```

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```

Status = LibLocateHandle(
    ByProtocol,
    &gEfiExtScsiPassThruProtocolGuid,
    NULL,
    &NoHandles,
    &HandleBuffer
);
if (EFI_ERROR(Status)) {
    BS->Exit(ImageHandle,EFI_SUCCESS,0,NULL);
}

//
// Loop through all the handles that support
// EFI_EXT_SCSI_PASS_THRU
//
for (Index = 0; Index < NoHandles; Index++) {

    //
    // Get the EFI_EXT_SCSI_PASS_THRU_PROTOCOL Interface
    // on each handle
    //
    BS->HandleProtocol(
        HandleBuffer[Index],
        &gEfiExtScsiPassThruProtocolGuid,
        (VOID **)&ExtScsiPassThruProtocol
    );

    if (!EFI_ERROR(Status)) {

        //
        // Use the EFI_EXT_SCSI_PASS_THRU Interface to
        // perform tests
        //
        Status = DoScsiTests(ScsiPassThruProtocol);
    }
}
return EFI_SUCCESS;
}

EFI_STATUS
DoScsiTests(
    EFI_EXT_SCSI_PASS_THRU_PROTOCOL *ExtScsiPassThruProtocol
)
{
    EFI_STATUS      Status;
    UINT32          Target;
    UINT64          Lun;
    EFI_EXT_SCSI_PASS_THRU_SCSI_REQUEST_PACKET Packet;
    EFI_EVENT       Event;

    //

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```
// Get first Target ID and LUN on the SCSI channel
//
Target = 0xffffffff;
Lun = 0;
Status = ExtScsiPassThruProtocol-> GetNextTargetLun(
    ExtScsiPassThruProtocol,
    &Target,
    &Lun
);

//
// Loop through all the SCSI devices on the SCSI channel
//
while (!EFI_ERROR (Status)) {

    //
    // Blocking I/O example.
    // Fill in Packet before calling PassThru()
    //
    Status = ExtScsiPassThruProtocol->PassThru(
        ExtScsiPassThruProtocol,
        Target,
        Lun,
        &Packet,
        NULL
    );

    //
    // Non Blocking I/O
    // Fill in Packet and create Event before calling PassThru()
    //
    Status = ExtScsiPassThruProtocol->PassThru(
        ExtScsiPassThruProtocol,
        Target,
        Lun,
        &Packet,
        &Event
    );

    //
    // Get next Target ID and LUN on the SCSI channel
    //
    Status = ExtScsiPassThruProtocol-> GetNextTargetLun(
        ExtScsiPassThruProtocol,
        &Target,
        &Lun
    );
}

return EFI_SUCCESS;
}
```